



BRAC UNIVERSITY
Department of Computer Science Engineering
Mid Term Examination, Summer 2023
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: **60 (20 x 3)**

Time: **90 Minutes**

- Total no. of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Use $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$
 $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.

Name:	ID:	Section:
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Question 1

[20]

Mr. Motlob is stranded on a deserted island with nothing but a couple of 5V batteries and a circuit. The circuit is shown in Figure 1. It consists of two diodes, several resistors, and a special bulb with a resistance of $100\text{k}\Omega$. The input voltages ' V_A ' and ' V_B ' can be either 0V or 5V. The special bulb has 3 working states:

1. Off: It emits no light (i.e. remains off) if the voltage across it is 0V.
2. Dim: It emits a dim light if the voltage across it is greater than 0V but less than 4V.
3. Bright: It emits a bright light if the voltage across it is greater than 4V.

As a cry for help, he plans to generate the sequence "Bright-Dim-Off-Bright" signal using this circuit.

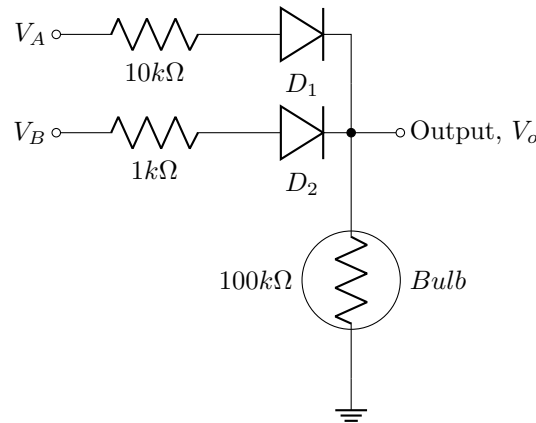


Figure 1

CO1	(a)	Solve the circuit to find the input voltages V_A and V_B for each of the instances of the signal sequence. You must show your calculations for finding the value of V_o . Copy table 1 and fill it up with your answers. <i>[Hint: Whenever any branch gets 5V as input, the diode of that particular branch will be On, otherwise it will be off]</i>	[12]
CO1	(b)	Calculate the average power dissipated in Watts for generating this signal. Assume each of the instances (Bright/Dim/Off/Bright) stays for an equal amount of time.	[8]

Table 1: Table for Question 1 (a)

State	V_o/V	V_A/V	V_B/V
Bright			
Dim			
Off			
Bright			

Question 2

[20]

For the RTL circuit in Figure 2, assume $\beta_F = 25$, $V_{BE(\text{saturation})} = 0.8\text{V}$, and $V_{CE(\text{saturation})} = 0.1\text{V}$.

CO1	(a)	Assume no loads are connected to driver RTL circuit. If so determine the logic gate that the driver circuit represents and justify your answer .	[6]
CO2	(b)	When both inputs are high ($V_A = V_B = 5\text{V}$) and no loads are connected to the driver, the total power dissipation in the driver circuit is 150 mW . Estimate the value of R_C in $k\Omega$.	[8]
CO2	(c)	Assume 7 load RTL inverters are connected to the output (V_o) of the driver circuit. Determine the output voltage (V_o) when both driver inputs are 0 V .	[6]

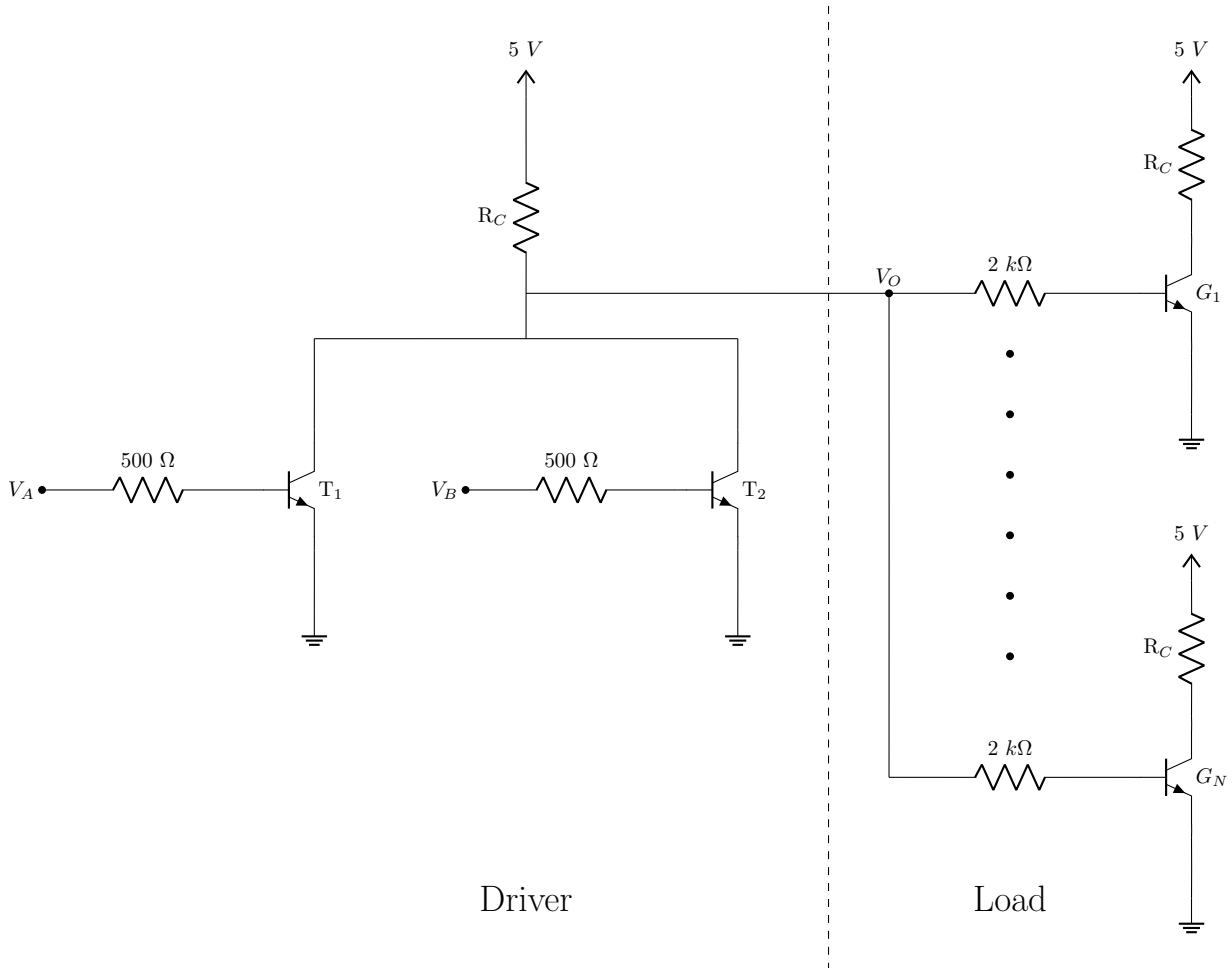


Figure 2

Question 3

[20]

For the DTL circuit in figure 3 assume $\beta_F = 65$, $V_{BE(saturation)} = 0.8V$, and $V_{CE(saturation)} = 0.2V$. The input voltages 'A' and 'B' can be either 0V or 5V.

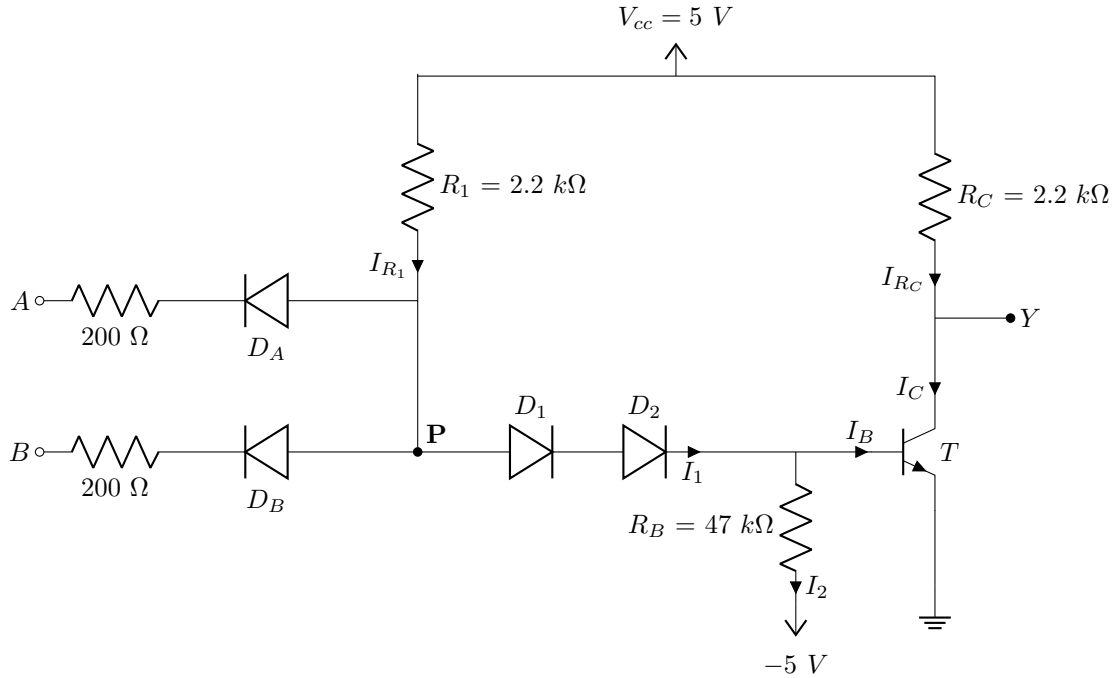


Figure 3

CO2	(a)	Determine the node voltage, P , for the DTL NAND gate when both the inputs 'A' and 'B' are LOW.	[4]
CO2	(b)	Identify what should be the minimum current gain, β_{min} to prevent the transistor from malfunctioning.	[7]
CO2	(c)	Assume that a designer wants to use the shown DTL NAND gate as a driver. The designer intends to drive multiple identical circuits from the Y-node of the driver. Estimate the highest number of gates the designer can drive without any reliability issues.	[9]

Question 4

[20]

For the given TTL circuit in figure 4, **common emitter current gain**, $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and **reverse common emitter current gain of the transistors** $\beta_R = 0.1$. Assume both inputs are **logical High** or **6V**.

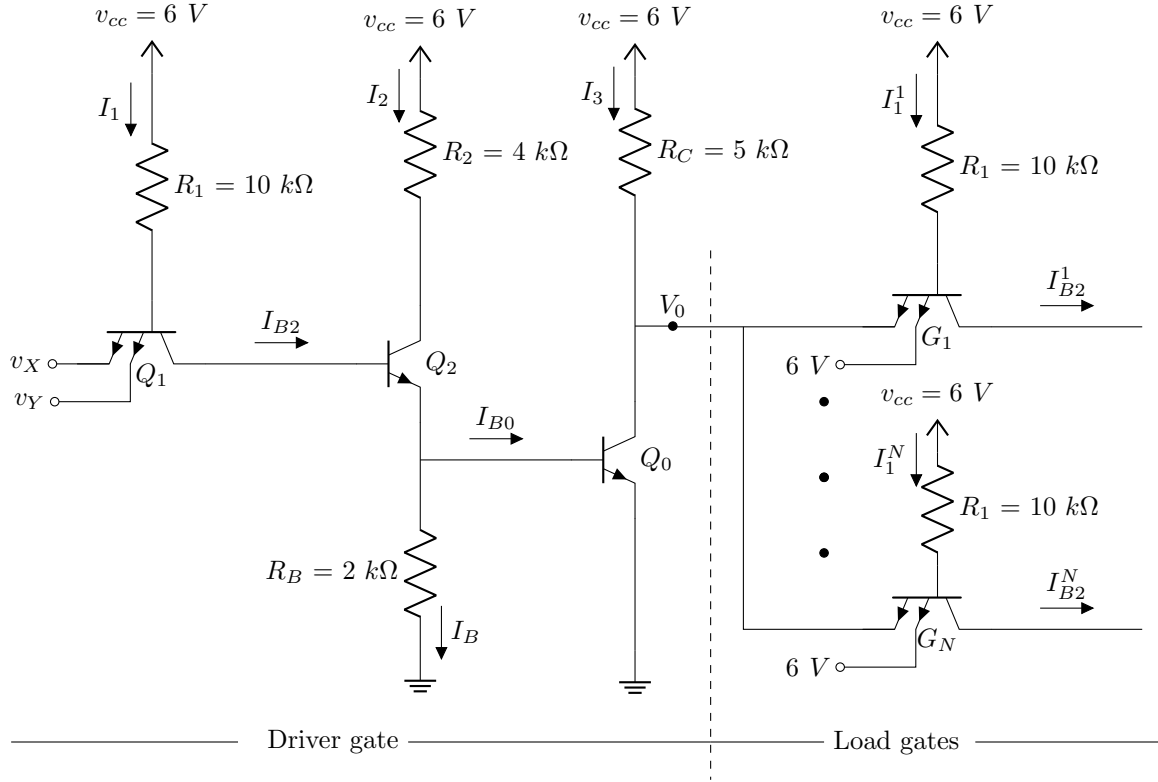


Figure 4

CO2	(a)	Determine the maximum possible fanout for the given input condition only .	[10]
CO2	(b)	Find the total power dissipation in the driver circuit if 10 load gates are connected to the driver.	[6]
CO2	(c)	Select a suitable minimum value of β_F for the BJT Q_0 if we want maximum possible fanout to be 70 .	[4]